

Marine microbiology

Lectures 7 – 8

**Microbial nutrition, cultivation
and influence of environmental
factors on growth**

Introduction

Synopsis (大纲) :

This lecture covers:

- factors affecting the culture of microbes
- media used in the culture of microbes
- environmental factors affecting microbial growth



B. subtilis colony.

Outcomes:

At the end of this lecture you will understand:

- the chemical composition of microbial cells
- the factors involved in cultivating microbes
- the types and general composition of media used to grow bacteria
- the influence of some major environmental factors on microbial growth, and the terminology (术语) used

Introduction

- Much of the study of microbes depends on being able to grow them, e.g.
 - to study structure, metabolism, physiology,
 - to prevent microbial growth where necessary
 - to diagnose pathogens and determine antibiotic sensitivity
 - to be able to use microbial biomass or products in commercial applications
- Only about 1% of bacteria in natural samples (soil, water) can be grown
 - evidence from comparison of 'viable' and 'total' counts
 - existence of VBNC - 'viable-but-non-culturable' bacteria
 - reflects our lack of knowledge of their requirements

Introduction

- Microbial cultivation requires knowledge of microbial requirements:
 - nutritional
 - environmental
e.g. temperature, osmotic effects, pH, oxygen, pressure
- **Nutrition** in microbes is influenced by:
 - I. sources of carbon, electrons and external energy used for growth
 - reflected in the diversity of nutritional types among microbes
 - II. the diversity of chemicals needed to make bacterial cell components
 - III. the need for growth factors
 - some microbes can make all their essential cell components from inorganic chemicals in the environment
 - others need to have the components they can't synthesize supplied (termed "growth factors")
 - amino acids, nucleic acids, vitamins

I - Sources of carbon, electrons and external energy

1. Carbon

- carbon forms the “backbone” of all organic molecules

2. Electrons

- electron flow in biological reactions generates energy
 - (e.g. the electron transport chain)
- electrons are needed to reduce molecules during biosynthesis
 - (e.g. the reduction of CO₂ to form organic carbon)

3. An external energy source

- necessary to power the cell (make ATP)

I - Sources of carbon, electrons and external energy

Principal requirement	Terminology for microbes involved
Carbon source	
1. CO ₂	Autotroph (自养生物) Greek <i>autos</i> = self and <i>trophe</i> = nutrition
2. Organic molecules made by other organisms	Heterotroph (异养生物) (Greek <i>heterone</i> = (an)other and <i>trophe</i> = nutrition)
Energy source	
1. Light	Phototroph (光能营养生物)
2. Exergonic chemical reactions (oxidation of organic or inorganic compounds)	Chemotroph (化能营养生物)
Electron source	
1. Reduced inorganic molecules	Lithotroph ("rock eater") 无机营养生物
2. Organic molecules	Organotroph 有机营养生物

Nutritional types among microbes based on carbon, electrons and external energy requirements



Cyanobacteria - photolithoautotrophs

Table 6.3 Major Nutritional Types of Microorganisms

<i>Nutritional Type</i>	<i>Carbon Source</i>	<i>Energy Source</i>	<i>Electron Source</i>	<i>Representative Microorganisms</i>
Photolithoautotroph	CO ₂	Light	Inorganic e ⁻ donor	Purple and green sulfur bacteria, cyanobacteria, diatoms
Photoorganoheterotroph	Organic carbon	Light	Organic e ⁻ donor	Purple nonsulfur bacteria, green nonsulfur bacteria
Chemolithoautotroph	CO ₂	Inorganic chemicals	Inorganic e ⁻ donor	Sulfur-oxidizing bacteria, hydrogen-oxidizing bacteria, methanogens, nitrifying bacteria, iron-oxidizing bacteria
Chemolithoheterotroph	Organic carbon	Inorganic chemicals	Inorganic e ⁻ donor	Some sulfur-oxidizing bacteria (e.g., <i>Beggiatoa</i>)
Chemoorganoheterotroph	Organic carbon	Organic chemicals, often same as C source	Organic e ⁻ donor, often same as C source	Most nonphotosynthetic microbes, including most pathogens, fungi, and many protists and archaea

Prescott's Microbiology 8th ed.

Photo / litho / autotroph
 Photo / organo / heterotroph
 Chemo / litho / autotroph
 Chemo / litho / heterotroph
 Chemo / organo / heterotroph

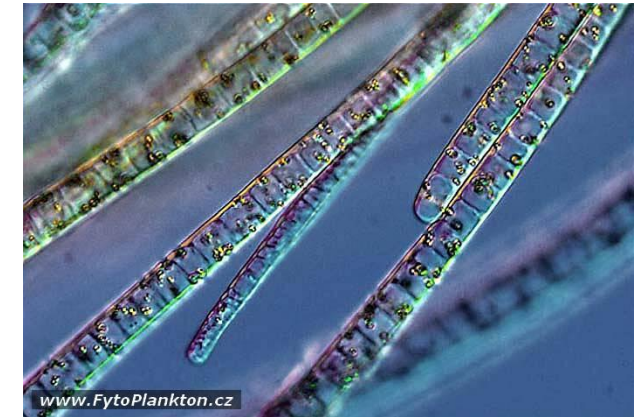
photo-光能
 chemo-化能
 litho-无机
 organo-有机

Nutritional types among microbes based on carbon, electrons and external energy requirements



Cyanobacteria 蓝藻细菌 - photolithoautotrophs

Nutritional Type	Carbon Source	Energy Source	Electron Source	Representative Microorganisms
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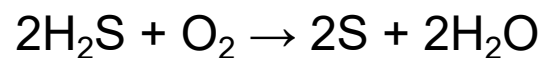
Beggiatoa 贝氏硫细菌

But some organisms have a mixed mode of nutrition (see Munn Table 3.1)

Mixotrophy = combination of autotrophic and heterotrophic processes, organism can switch between them

■ *Beggiatoa*: combination of lithoautotrophy and organoheterotrophy

- Can oxidise (氧化) organic compounds to give CO₂, but some species can also oxidise hydrogen sulphide (硫化氢) for energy



■ Some flagellates (鞭毛虫) and dinoflagellates (鞭毛藻): combination of both photoautotrophy (light and CO₂) and organoheterotrophy (engulf prey 吞食猎物 such as bacteria)

II -The diversity of chemicals needed to make microbial cell components

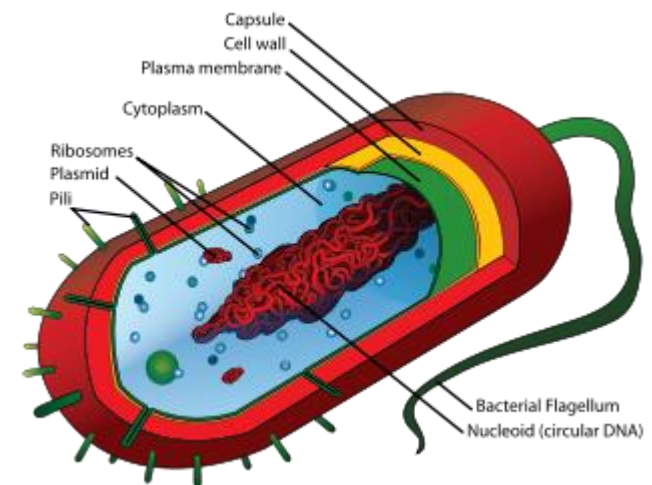
The major components in a bacterial cell (*Escherichia coli*):

WATER **70%**

DRY WEIGHT **30%**

Composed of (approximately):

DNA	3%
RNA	12%
Protein	70%
Polysaccharide	5%
Lipid	6%
Phospholipid	4%
Other	<1%



II - The diversity of chemicals needed to make microbial cell components

Nutrient (macronutrient)	Examples of requirement	Major external sources
Carbon	Most cell structures (e.g. cell walls) and functional chemicals (e.g. DNA)	CO ₂ (inorganic) Simple and complex sugars (organic)

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Nitrogen	Proteins, nucleic acids, some polysaccharides and lipids	NH ₄ ⁺ , NO ₃ , N ₂ (inorganic) Proteins, amino acids, urea (organic)

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Hydrogen	All organic compounds	Organic compounds, H ₂ O
Oxygen	Many organic compounds Electron transport chain (respiration 呼吸作用)	H ₂ O, CO ₂ , O ₂

II - The diversity of chemicals needed to make microbial cell components

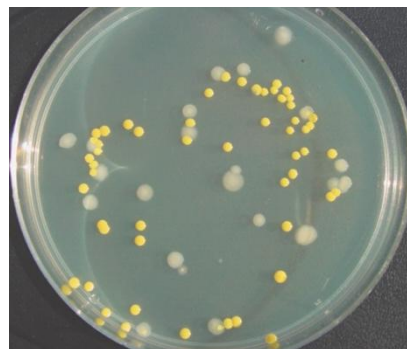
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Hydrogen	All organic compounds	Organic compounds, H ₂ O
Oxygen	Many organic compounds Electron transport chain (respiration)	H ₂ O, CO ₂ , O ₂
Sulphur	S-amino acids (methionine-甲硫氨酸, cysteine-半胱氨酸) Vitamins (biotin 生物素 H, thiamine 硫胺素 B1)	SO ₄ (inorganic) cysteine, methionine (organic)
Phosphorus	Nucleic acids, phospholipids, ATP	PO ₄

II - The diversity of chemicals needed to make microbial cell components

Nutrient (macronutrient , micronutrient)	Examples of requirement	Major external sources
Carbon	Most cell structures (e.g. cell walls) and functional chemicals (e.g. DNA)	CO ₂ (inorganic) Simple and complex sugars (organic)
Nitrogen	Proteins, nucleic acids, some polysaccharides and lipids	NH ₄ ⁺ , NO ₃ , N ₂ (inorganic) Proteins, amino acids, urea (organic)
Hydrogen	All organic compounds	Organic compounds, H ₂ O
Oxygen	Many organic compounds Electron transport chain (respiration)	H ₂ O, CO ₂ , O ₂
Sulphur	S-amino acids (methionine, cysteine) Vitamins (biotin, thiamine)	SO ₄ (inorganic) cysteine, methionine (organic)
Phosphorus	Nucleic acids, phospholipids, ATP	PO ₄
K, Ca, Mg, Fe, Mn, Zn, Co, Ni, Cu	Typically cofactors(辅因子) ('helper molecules'; bind to and assist enzymes to function properly)	As ions (e.g. K ⁺) in solution

III - Growth factor requirements in microbes

- Organic compounds that are essential for growth but cannot be synthesized by the organism. Includes:
 - the 20 essential amino acids
 - the purine (嘌呤) and pyrimidine (嘧啶) bases for making nucleic acids
 - vitamins
- **All autotrophs** and some **heterotrophs** can synthesise all their growth factor requirements
 - they have complete sets of genes for all biosynthetic pathways.
- Some bacteria need to be supplied with one, several or most growth factors
 - BY4741 MATa his3 Δ 1 leu2 Δ 0 met15 Δ 0 ura3 Δ 0



Uptake of nutrients by microbes

Problems for the cell:

- microbes can only take in dissolved (溶解的) molecules - they must take in nutrients from dilute solutions (稀溶液) into the cell against a concentration gradient (浓度梯度)
- cytoplasmic membrane prevents passage of most substances (物质)
- there is an enormous variety of nutrients
- nutrient uptake mechanisms should be specific

Bacteria have developed a number of mechanisms for nutrient uptake across the membrane:

1. passive diffusion 被动扩散

- small molecules only e.g. O_2 , H_2 , H_2O

2. facilitated diffusion 易化扩散 – ATP not required

3. active transportation 主动运输 – ATP required

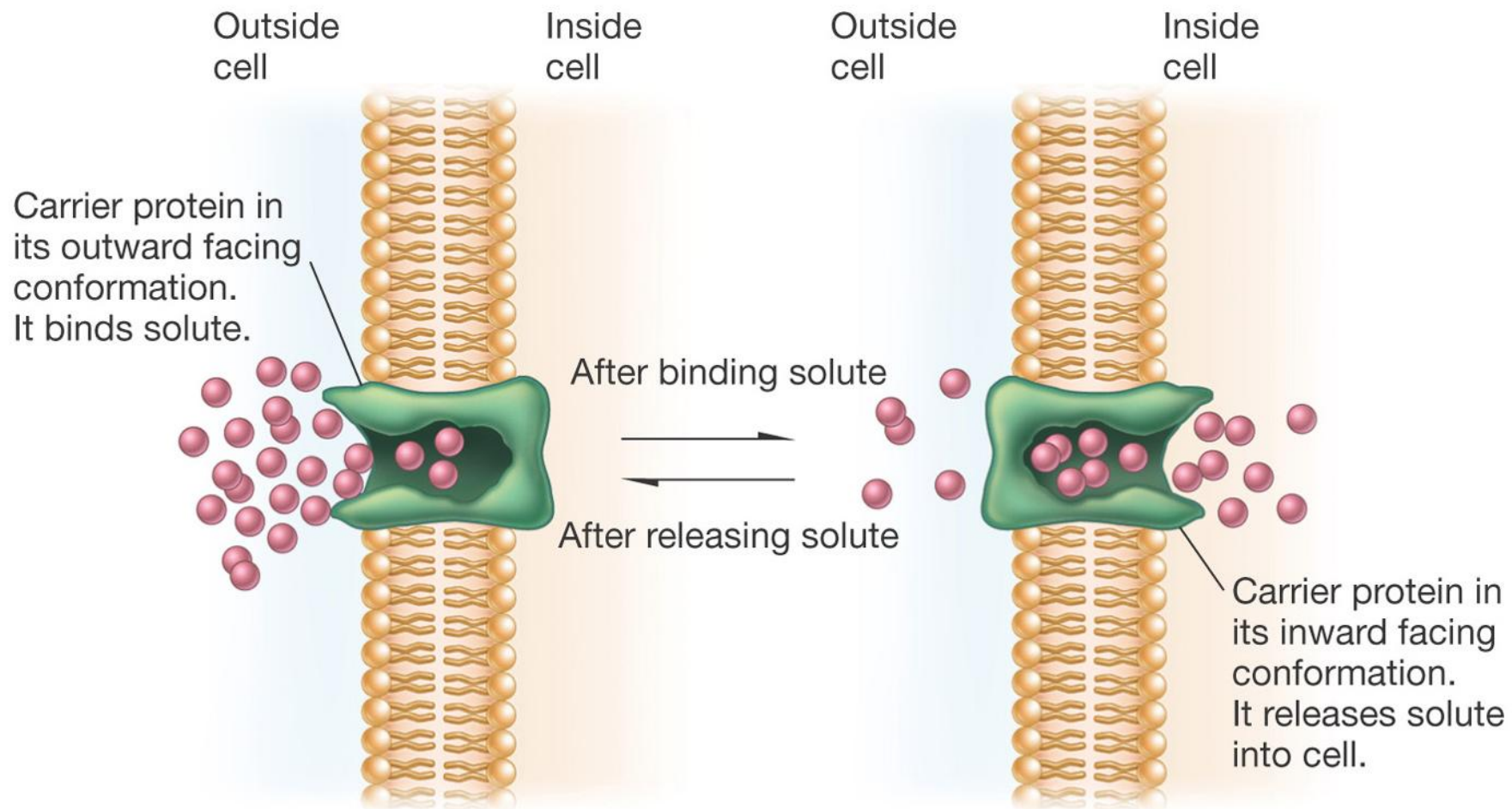
4. group transportation 基团运输 – ATP required

- Transport of solute molecules to higher concentrations (against a concentration gradient) in which the molecule is chemically-modified during transport

Uptake of nutrients by bacteria

Facilitated diffusion

- Diffusion across the membrane involving carrier proteins
- ATP energy not required

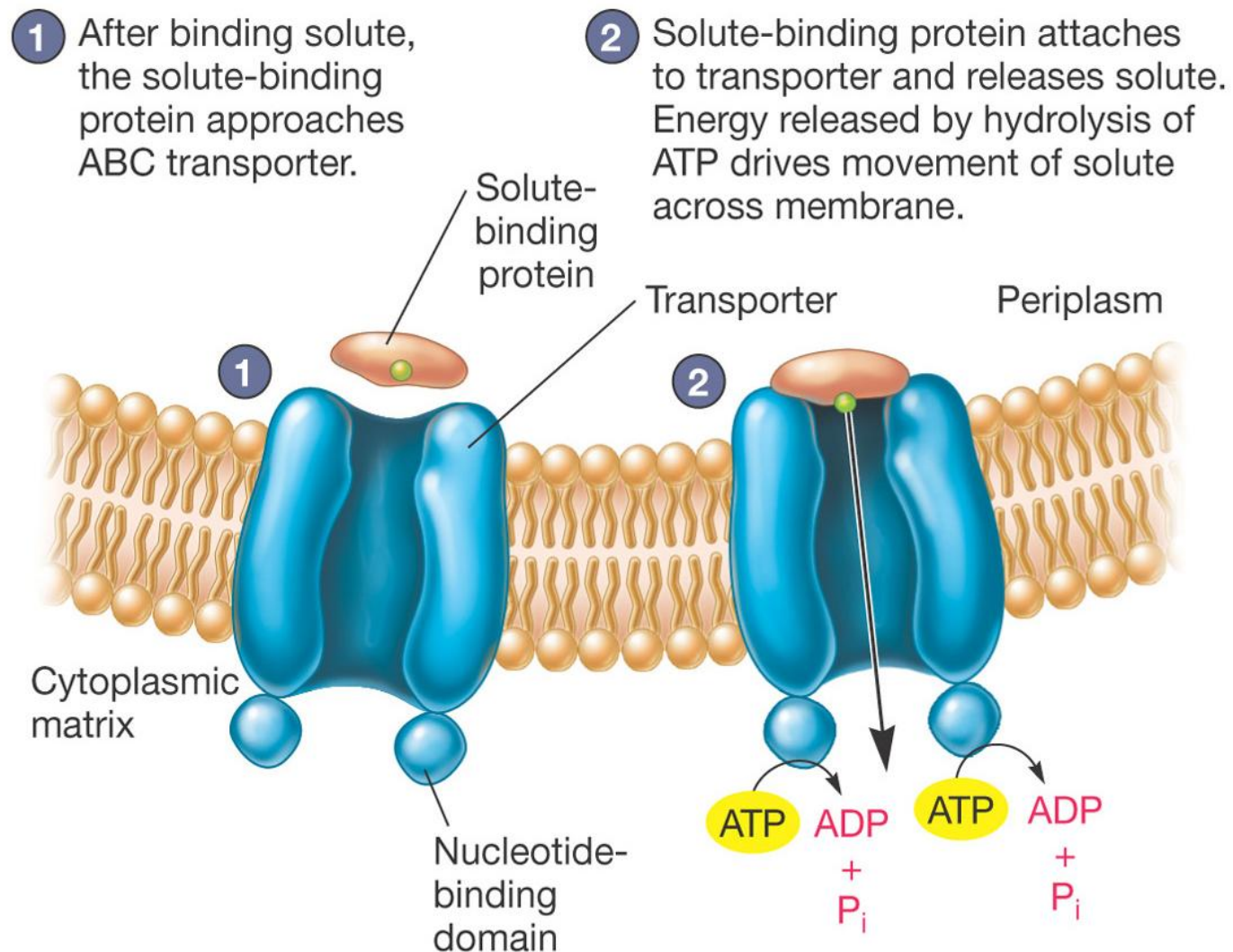


Uptake of nutrients by bacteria

Active transport

- Transport of solute molecules to higher concentrations (against a concentration gradient) using transport proteins
- ATP energy is required

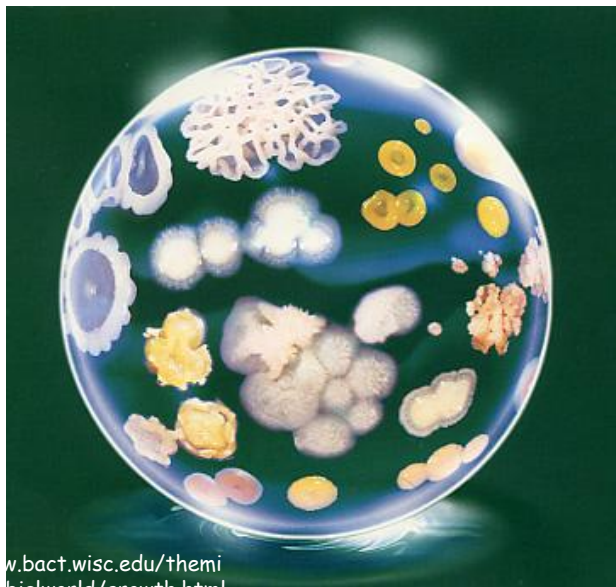
One example:
ABC Transporters



Culture Media (media = plural; medium = singular)



- The design of culture media for microbes is determined by:
 - I. sources of carbon, electrons and external energy used for growth
 - II. the diversity of chemicals needed to make microbial cell components
 - III. the need for growth factors



- **Two common methods**

1. **Liquid culture**

Microbes grown as a liquid suspension ('broth肉汤 culture')

2. **Solid culture**

Microbes grown on an **agar** (gel) surface ('plate culture')

w.bact.wisc.edu/themes/microbialworld/growth.html

Culture Media

Agar

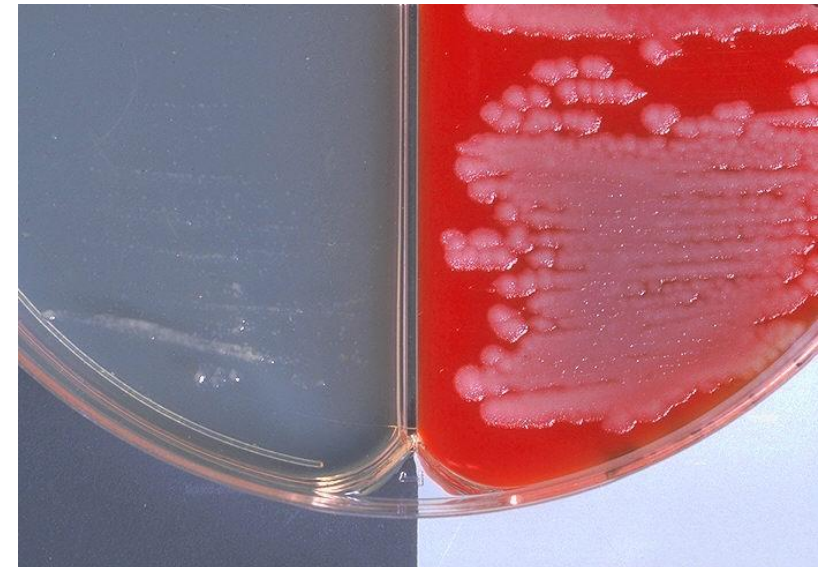
- a **carbohydrate**, extracted from red algae
- usually added ($\sim 1\text{--}2\%$) to a liquid broth medium

Agar properties:

- **dissolves** @ 100°C , **sets** @ $\sim 40^\circ\text{C}$, stable from 0°C to $>80^\circ\text{C}$ when set
- easily **sterilised** (autoclaving 121°C , 15kPa)
- **not degraded** by bacteria (except for a couple of species)
- Spread plate (涂布), streak plate (划线) or pour plate yields single isolated **colonies**



The red alga *Hypnea* sp.



Types of Culture Media

Media such as tryptic（胰蛋白胨）soy broth or agar are called **general purpose media**（通用培养基）because they support the growth of many micro-organisms. Blood and other special nutrients may be added to general purpose media to encourage the growth of fastidious（需要复杂营养的）heterotrophs (**enriched media**富集培养基)

Types of media include:

1. **Defined or Synthetic** 限定或合成培养基
2. **Complex** 天然培养基
3. **Selective** 选择培养基
4. **Differential** 鉴别培养基
5. **Selective and Differential**

Types of Culture Media

1. Defined or Synthetic media

- A medium in which all components are known
 - Concentration of each ingredient is known precisely (确切地) : inorganic salts + simple organic compounds
- Consistency (一致) from batch to batch, therefore used widely in research, as it is often desirable (令人满意的) to know what the experimental organism is metabolising.
- Useful for a **small** range of microbes (requires precise knowledge of growth requirements of the organism to be cultured)
 - e.g. *E. coli* minimal medium contains
 - a carbon source - glucose
 - a nitrogen source - $(\text{NH}_4)_2\text{SO}_4$
 - various sulfates, phosphates, trace (微量) metals

Types of Culture Media

2. Complex media

- A medium containing some ingredients of unknown chemical composition
 - e.g. peptone (蛋白胨) and yeast extract (酵母膏)
 - **peptones** are prepared by proteolytic (蛋白水解) digestion of meat, casein (酪蛋白), soya meal (豆渣), gelatin (明胶) and other protein sources (a source of amino acids and small peptides)
 - **yeast extract** is a lysate (溶解产物) of yeast cells (a source of vitamins and other growth factors)
- Formulated to support the growth of a **wide** range of microbes
- Used where the nutritional requirements of a particular organism are unknown
 - e.g. many fastidious bacteria with complex nutritional or cultural requirements e.g. *Neisseria*.

Examples of complex media

Nutrient Broth/Agar

Peptone (gelatin hydrolysate) 牛肉膏蛋白胨培养基
Beef extract

Tryptic Soy Broth/Agar (TSB/TSA) 胰蛋白胨大豆培养基

Tryptone (pancreatic胰腺的 digest of casein)
Peptone (soybean hydrolysate)
Glucose
NaCl
K₂HPO₄

MacConkey Agar 麦氏培养基

Pancreatic digest of gelatin
Pancreatic digest of casein
Lactose (乳糖)
NaCl
Bile salts (胆盐)
Neutral red (中性红)
Crystal violet (结晶紫)
Agar

Marine Broth/Agar

Artificial seawater medium

- Artificial seawater(ASW) is a mixture of dissolved mineral salts that simulates (模拟) seawater.
- ASW allows the easy preparation of media suitable for marine microbes
- It has the advantage over natural seawater of reproducibility (可再生性)
- There are various recipes for ASW – one basic recipe includes:
 - NaCl
 - MgSO₄
 - MgCl₂
 - CaCl₂
 - KCl
 - NaHCO₃



Artificial Sea Water Salts

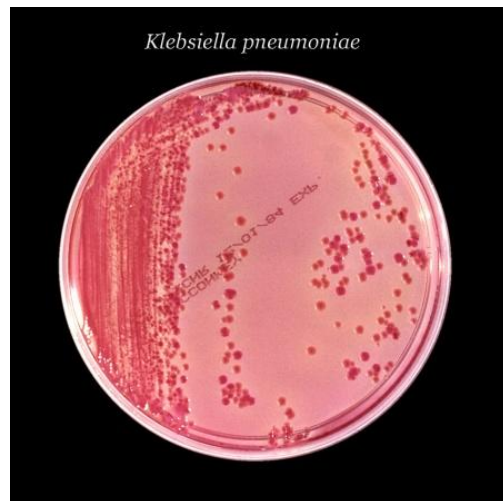
Marine agar and marine broth are complex media

- A complex carbon source is added to ASW – e.g. one or two of the following
 - Beef extract
 - Peptone
 - Yeast extract

Types of Culture Media

3. Selective media

- Have ingredients (成分) /conditions which **inhibit** the growth of **unwanted** microbes, and **favour** the growth of **specific** organisms.
- e.g. MacConkey agar uses **bile salts** and the dye **crystal violet**, to inhibit Gram +ve but not Gram –ve bacteria
- e.g. Addition of **sodium azide** (NaN_3) inhibits Gram –ve but not Gram +ve bacteria
- e.g. Addition of **salt** (7.5%) to Nutrient agar as in Mannitol (甘露糖) salt agar selects for growth of salt-tolerant bacteria such as staphylococci (葡萄球菌) from skin. Inhibits other bacteria unable to grow in high salt.



Types of Culture Media

3. Selective media

- Bacteria may also be selected by incubation with nutrients that they specifically use
 - e.g. **pesticide-degrading bacteria**
 - the pesticide (杀虫剂) is supplied as the sole (唯一) C source available for growth
 - only those bacteria capable of metabolizing the pesticide will grow
 - used to enrich and isolate (分离) rare microorganisms



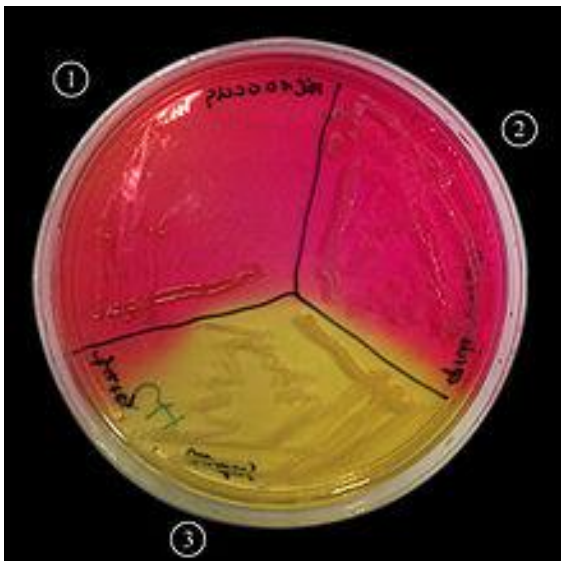
www2.dijon.inra.fr/umrmse/spip.php?article30

Types of Culture Media

4. Differential media

- Distinguish between **different genera** or **species** based on their biological / metabolic properties. Often contain acid-base indicators (指示剂) and selective agents to limit the range of bacteria growing.
- Many media are both selective and differential

e.g. **MacConkey agar**: has **lactose** and **neutral red dye**; lactose-fermenting (乳糖发酵) colonies (*E. coli*) are red while non-fermenters (*Salmonella* 沙门氏菌) are colourless.



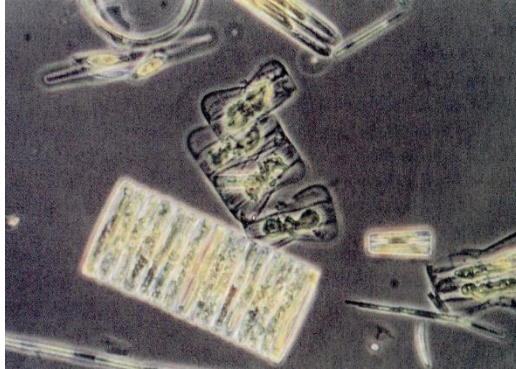
e.g. **Mannitol Salt agar**: *Staphylococcus aureus* (金黄色葡萄球菌) ferments **mannitol** producing acids, **pH indicator** goes from red to yellow. Non-pathogenic staphylococci remain red.

Environmental factors that affect microbial growth

1. Temperature



www.nsf.gov/pubs/1997/antpanel/5signif.htm



www.bact.wisc.edu/themicrobialworld/intro.htmlif.htm

- Varies with latitude (纬度), vertically (垂直地), and over small to global scales

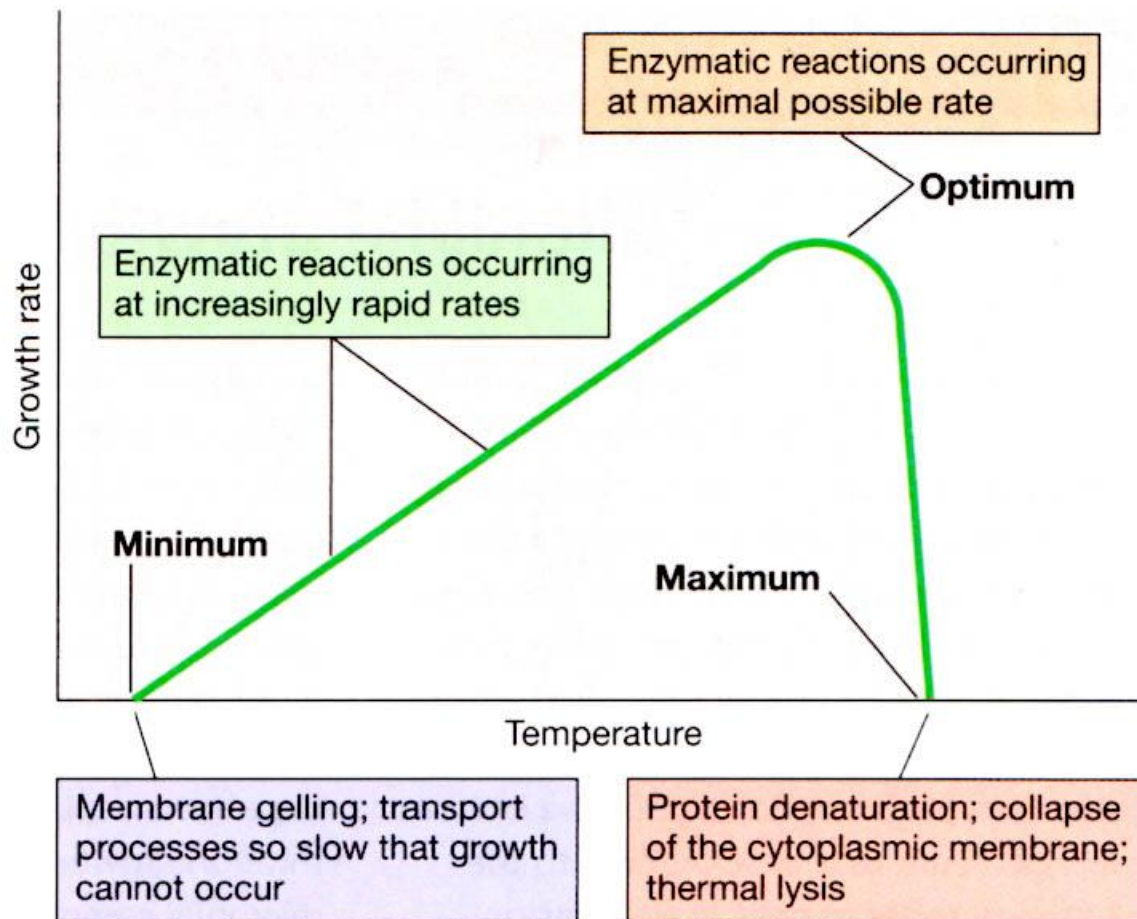
Extremes include:

- Seawater below polar ice -10°C
- Hydrothermal vents (深海热泉) $+110^{\circ}\text{C}$
- Thermal springs (温泉) 100°C
- Many bacteria/archaea thrive at these extremes



<http://oceanexplorer.noaa.gov/>

1. Temperature



Cardinal temperatures

- The **minimum**, **optimum** (最适宜) and **maximum** temperatures for bacterial growth
- For any bacterium, the range from minimum to maximum temperature is generally 35 to 40°C
 - some species, e.g. pathogens, can have a much narrower ranges

- from **minimum** to **optimum** – metabolic reactions accelerate and growth rate increases
- **at optimum** – cellular reactions are optimum
- **from optimum** to **maximum** – structure (e.g. lipids melt, proteins denature) and function disrupted, cell function ceases

1. Temperature

Bacteria divide into 5 groups based on temperature growth range

1. Psychrophiles (嗜冷菌): $<1 - <20^{\circ}\text{C}$: Grow well at 0°C

e.g. *Moritella* sp. (optimum 9°C) from deep ocean, Korea

2. Psychrotrophs (耐冷菌): $0 - 35^{\circ}\text{C}$

optima in mesophilic range but able to grow at $0-7^{\circ}\text{C}$, common food spoilage (腐败) organisms

3. Mesophiles (嗜温菌) : $15 - 44^{\circ}\text{C}$

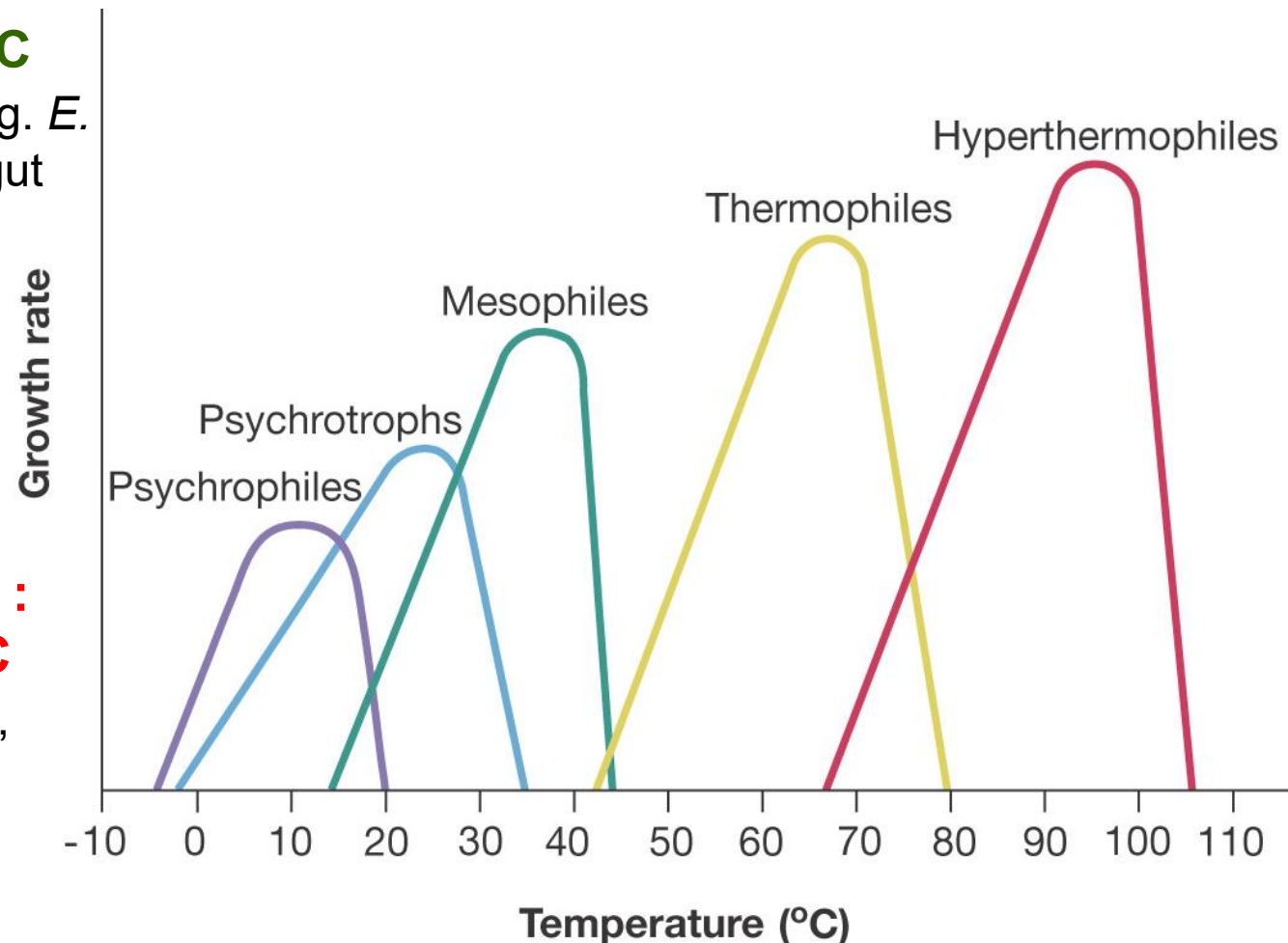
moderate (温和) temperatures e.g. *E. coli* (optimum 37°C), from human gut

4. Thermophiles (嗜热菌) : $42 - 80^{\circ}\text{C}$: optima $55 - 60^{\circ}\text{C}$

grow best at high temps e.g. hot springs

5. Hyperthermophiles (超嗜热菌) : $70 - >100^{\circ}\text{C}$: optima $85 - \sim 113^{\circ}\text{C}$

e.g. *Pyrolobus fumarii* (opt. 106°C), hydrothermal vent, mid Atlantic



2. Osmotic effects/water availability

- Most microbial activity occurs in free water or at high humidity (湿度)
- Essential for life - biochemical processes in cells occur in an aquatic (fluid) environment
- Can contain low to high concentrations of solutes (salts and other inorganic and organic chemicals)
 - **Hypotonic (低滲) environment:** Osmotic concentration inside the cell > outside the cell
 - **Hypertonic (高滲) environment:** Osmotic concentration inside the cell < outside the cell
 - Most marine and salt lake bacteria live in hypertonic environments
 - Some microbes are adapted to extreme hypertonic environments e.g. Dead Sea

2. Osmotic effects/water availability

Halotolerance/Osmotolerance

- The osmotic concentration of most hypertonic environments is a consequence of NaCl and other salts
- hence bacteria are characterized according to their 'halo(salt)tolerance'

Nonhalophile (非喜盐)

- no salt requirement for growth
- salt inhibitory at low conc. (< 2%)

Halotolerant (耐盐)

- no salt requirement for growth
- able to tolerate moderate salt concs

Halophilic/halophile (喜盐)

- require salt for growth
 - mild halophiles (1-6%)
 - moderate halophiles (7-15%)

Extremely halophilic/extreme halophile

- require 15-30% salt for growth

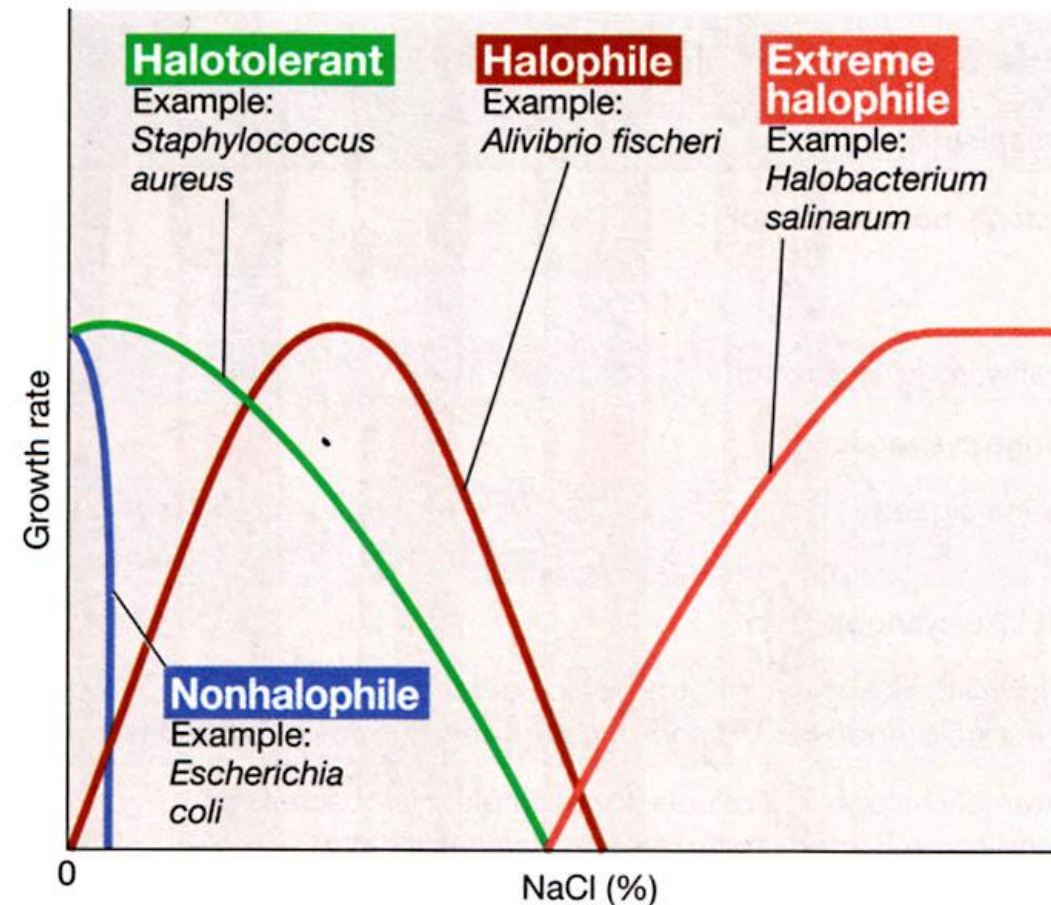
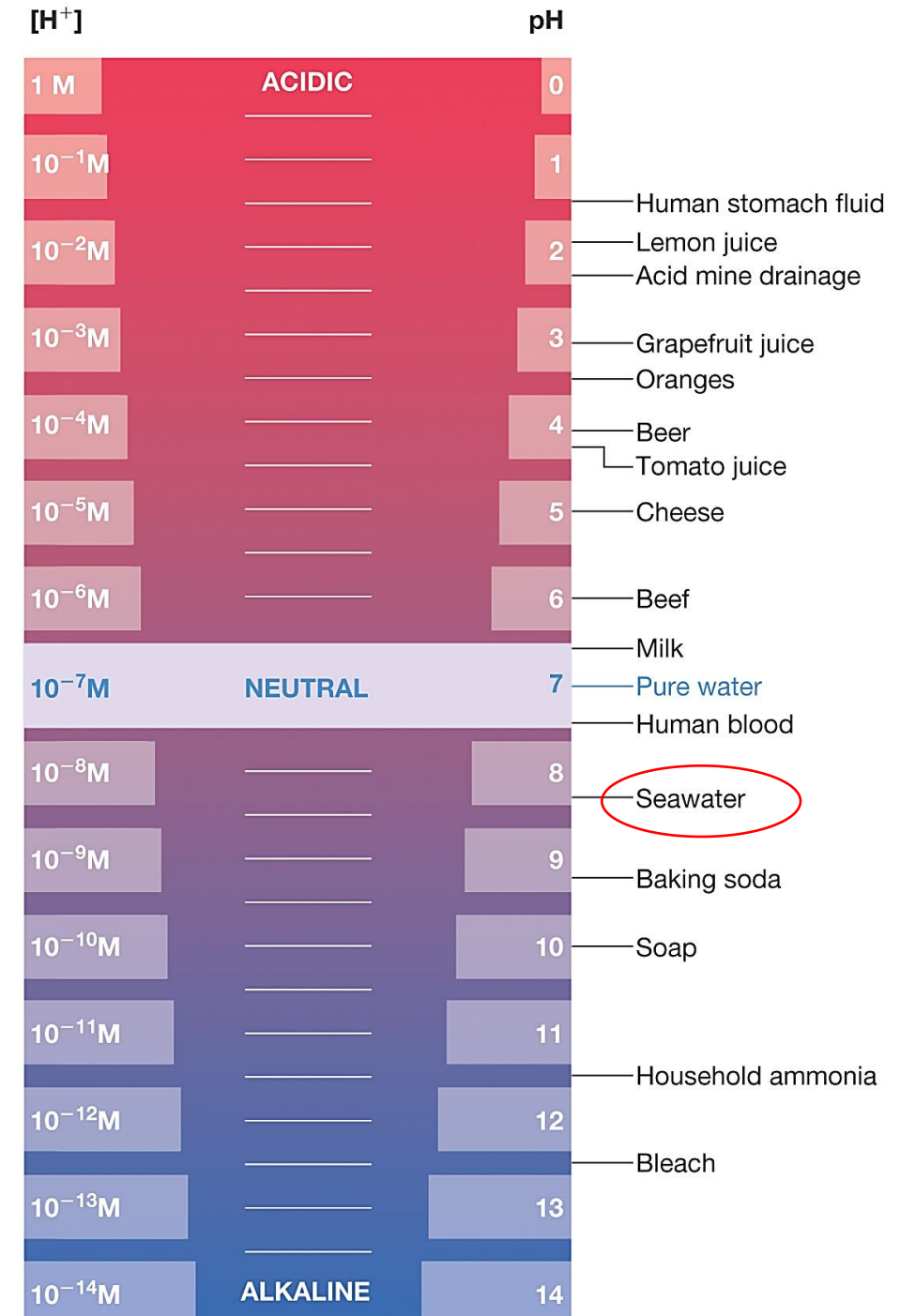


Figure 6.25 Effect of sodium chloride concentration on growth of microorganisms of different salt tolerances or requirements.

3. pH and its effects on microbes

- is a measure of the $[H^+ \text{ ion}]$
- reflects the relative acidity/alkalinity 酸/碱性 of the environment
- ranges from 0–14
 - a logarithmic (对数的) scale: each unit change is a 10-fold change in $[H^+ \text{ ion}]$
- varies enormously among habitats (栖息地)
 - gastric juices (胃液)/acid thermal springs: pH < 1
 - alkaline soils/soda lakes: pH 9–11
- bacteria differ markedly in their tolerance of pH conditions
 - most live in the range pH 4–9
 - majority are “neutrophiles 嗜中性” (pH 5.5–8)
 - relatively few grow at < 3 or > 9
 - seawater pH ~8, most marine microbes “alkaliphiles 嗜碱性”
- Most fungi prefer pH 4 – 6
- Many archaea are “acidophiles 嗜酸性” e.g. pH 0-3



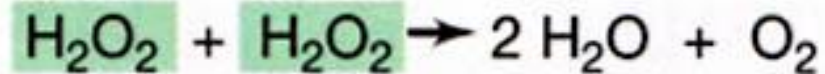
4. Oxygen

- the Earth's atmosphere is about 20% O₂
- many ecosystems and habitats are anaerobic (no O₂)
 - swamps(沼泽), the rumen (瘤胃cow's stomach), gastrointestinal tracts(消化道), anoxic(缺氧的) zone in oceans
- microbes differ markedly in tolerance to O₂
- differing tolerances may reflect evolution from an O₂-free planet

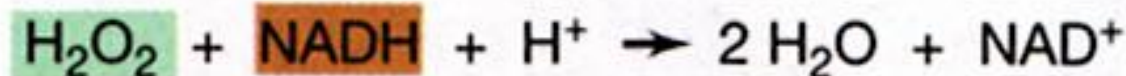
4. Oxygen

Oxygen toxicity

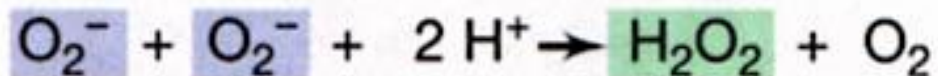
- O_2 or O_2 - products are very reactive
- Many O_2 - products are toxic:
 - superoxide (O_2^-)
 - hydrogen peroxide (H_2O_2)
 - hydroxyl ($-OH$) radicals and singlet oxygen (O^-)
- Toxic O_2 - products can damage cells by oxidising proteins, lipids and DNA
- Microbes that tolerate O_2 have enzymes that protect against these toxic products



(a) **Catalase**



(b) **Peroxidase**

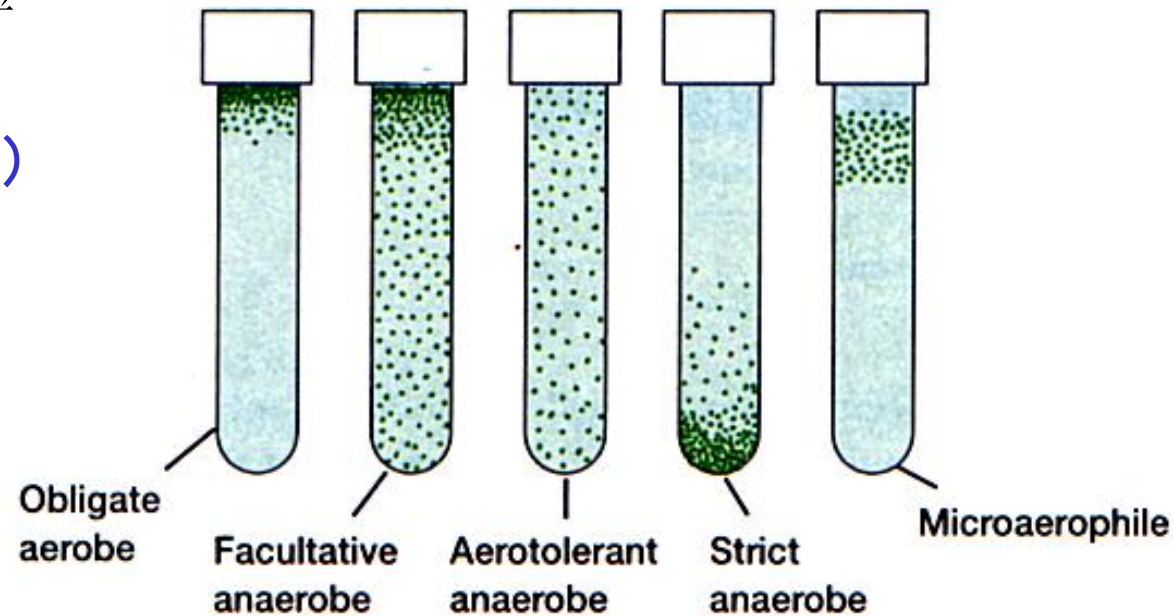


(c) **Superoxide dismutase**

4. Oxygen

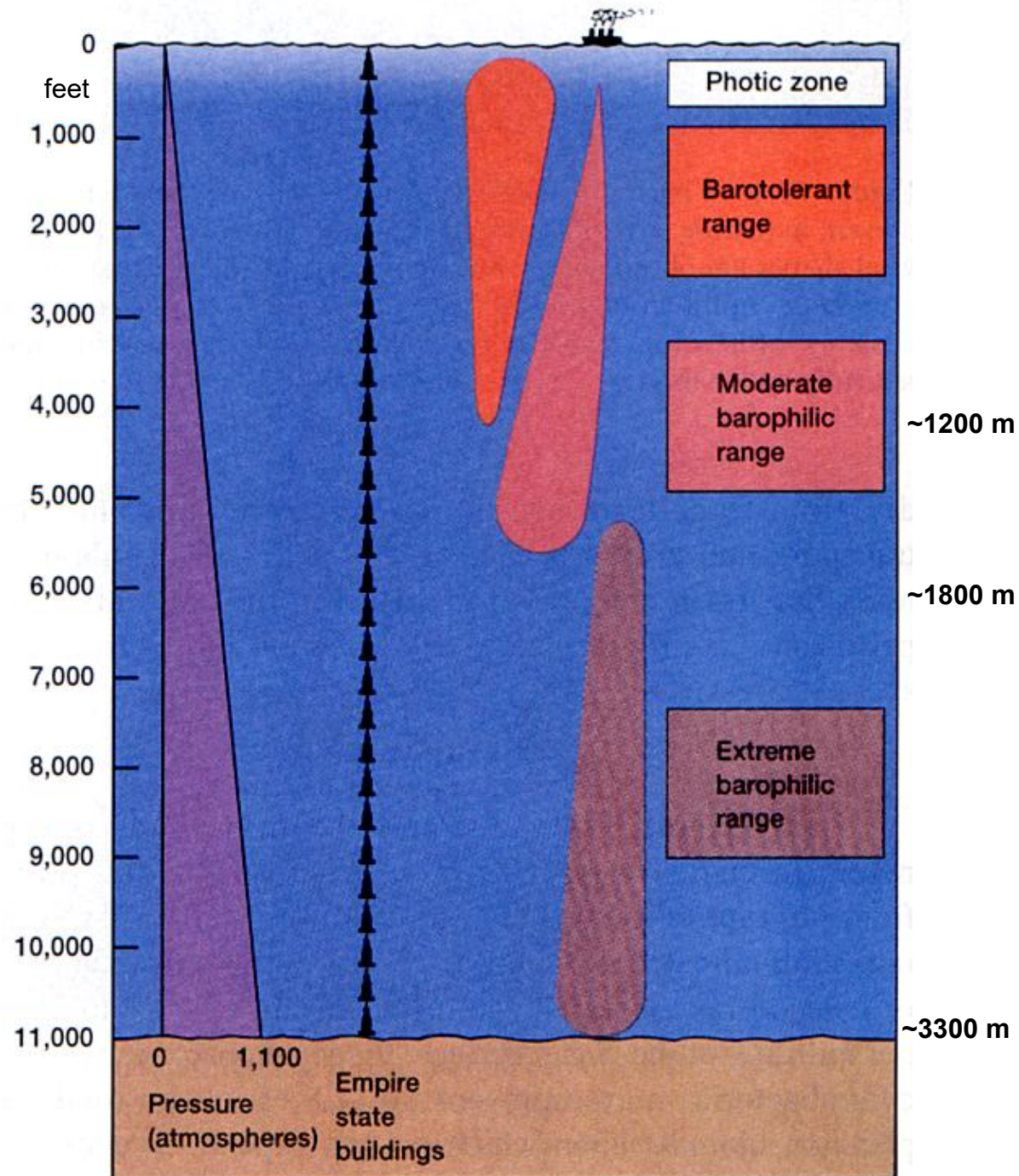
- **Obligate aerobes (专性需氧)**
 - completely dependent on atmospheric O_2 for growth
 - gain energy by aerobic respiration (呼吸作用) *
- **Obligate/Strict anaerobes (专性厌氧)**
 - killed in the presence of O_2
 - anaerobic respiration or fermentation
- **Facultative anaerobes (兼性厌氧)**
 - do not require O_2 for growth but grow better in its presence
 - aerobic respiration or fermentation
- **Aerotolerant anaerobes (耐氧性厌氧)**
 - grow equally well with or without O_2
 - fermentation
- **Microaerophiles (微需氧)**
 - damaged by normal atmospheric levels of O_2 , require 2-10% O_2

Microbes divide into 5 groups based on O_2 tolerance



5. Pressure

- In ocean depths of 1000 m or more, hydrostatic pressure (流体静压) can reach 600 – 1100 atmospheres. Only **barophilic** (嗜压) and **barotolerant** (耐压) bacteria and archaea grow
- High pressure disrupts DNA and protein synthesis, membrane fluidity/transport, enzyme activity
- **Barophiles (piezophiles)** require high pressures for maximal growth
 - one from Mariana trench (10,500 m) only grows at 400-500 atm at 2°C
 - **Hyperpiezophiles** grow best at > 590 atm
- **Barotolerant** organisms can grow at high pressures but do not require them,
 - increased pressure adversely affects them, but not as much as non-tolerant microbes



Summary

- **The ability to grow microbes is fundamental to microbiology**
- **Factors in microbial nutrition include sources of carbon, energy and electrons; cell chemical composition; and the need for growth factors**
- **There are four mechanisms by which bacteria take up nutrients across the plasma membrane**
- **Culture methods include the use of liquid and solid media**
- **Culture media include defined, complex, selective, differential and selective/differential media**
- **Environmental factors affecting microbial growth in general include temperature, osmotic effects, pH, O₂ and pressure**

Reference:

Prescott's Microbiology. Chapter 7

